



SMDM PROJECT

Business report



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Great Learning

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1. Wholesale Customers Analysis

Problem Statement:

A wholesale distributor operating in different regions of Portugal has information on annual spending of several items in their stores across different regions and channels. The data consists of 440 large retailers' annual spending on 6 different varieties of products in 3 different regions (Lisbon, Oporto, Other) and across different sales channel (Hotel, Retail).

1.1 Use methods of descriptive statistics to summarize data. Which Region and which Channel spent the most? Which Region and which Channel spent the least?

Descriptive statistics is concerned with Data Summarization Graphs/Charts and tables. The methods of descriptive statics include Distribution, which deals with each value's frequency, Measures of Central Tendency and Measures of variability. The most widely used measures of central tendency is Arithmetic Mean, Median, and Mode.

Mean is defined as the arithmetic average of all observations in the data set.

Median is defined as the middle value in the data set arranged in ascending or descending order.

Mode is defined as the most frequently occurring value in the distribution; it has the largest frequency.

Measures of Dispersion include Range, IQR, Standard Deviation

Range is the simplest of all measures of dispersion. It is calculated as the difference between maximum and minimum value in the data set.

Inter-Quartile Range (IQR) is computed on middle 50% of the observations after eliminating the highest and lowest 25% of observations in a data set that is arranged in ascending order. IQR is less affected by outliers.

Standard deviation is the square root of variance in simple words

The table below shows the description of the Wholesale customer dataset:

	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicatessen
count	440.00	440.00	440.00	440.00	440.00	440.00
mean	12000.30	5796.27	7951.28	3071.93	2881.49	1524.87
std	12647.33	7380.38	9503.16	4854.67	4767.85	2820.11
min	3.00	55.00	3.00	25.00	3.00	3.00
25%	3127.75	1533.00	2153.00	742.25	256.75	408.25
50%	8504.00	3627.00	4755.50	1526.00	816.50	965.50
75%	16933.75	7190.25	10655.75	3554.25	3922.00	1820.25
max	112151.00	73498.00	92780.00	60869.00	40827.00	47943.00

In the table below we can see some sample records which has 2 categorical variable and 6 numerical variables. The data consists of 440 large retailers' annual spending on 6 different varieties of products in 3 different regions (Lisbon, Oporto, Other) and across different sales channel (Hotel, Retail).

	Buyer/Spender	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicatessen
0	1	Retail	Other	12669	9656	7561	214	2674	1338
1	2	Retail	Other	7057	9810	9568	1762	3293	1776
2	3	Retail	Other	6353	8808	7684	2405	3516	7844
3	4	Hotel	Other	13265	1196	4221	6404	507	1788
4	5	Retail	Other	22615	5410	7198	3915	1777	5185

	Buyer/Spender	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicatessen
435	436	Hotel	Other	29703	12051	16027	13135	182	2204
436	437	Hotel	Other	39228	1431	764	4510	93	2346
437	438	Retail	Other	14531	15488	30243	437	14841	1867
438	439	Hotel	Other	10290	1981	2232	1038	168	2125
439	440	Hotel	Other	2787	1698	2510	65	477	52

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 440 entries, 0 to 439
Data columns (total 9 columns):
Buyer/Spender      440 non-null int64
Channel            440 non-null object
Region             440 non-null object
Fresh              440 non-null int64
Milk               440 non-null int64
Grocery            440 non-null int64
Frozen             440 non-null int64
Detergents_Paper   440 non-null int64
Delicatessen       440 non-null int64
dtypes: int64(7), object(2)
memory usage: 31.1+ KB

```

	Delicatessen	Detergents_Paper	Fresh	Frozen	Grocery	Milk
Channel						
Hotel	1415.96	790.56	13475.56	3748.25	3962.14	3451.72
Retail	1753.44	7269.51	8904.32	1652.61	16322.85	10716.50

	Delicatessen	Detergents_Paper	Fresh	Frozen	Grocery	Milk
Region						
Lisbon	1354.9	2651.12	11101.73	3000.34	7403.08	5486.42
Oporto	1159.7	3687.47	9887.68	4045.36	9218.60	5088.17
Other	1620.6	2817.75	12533.47	2944.59	7896.36	5977.09

```

Region
Lisbon      2386813
Oporto       1555088
Other       10677599
Name: Spending, dtype: int64

```

```

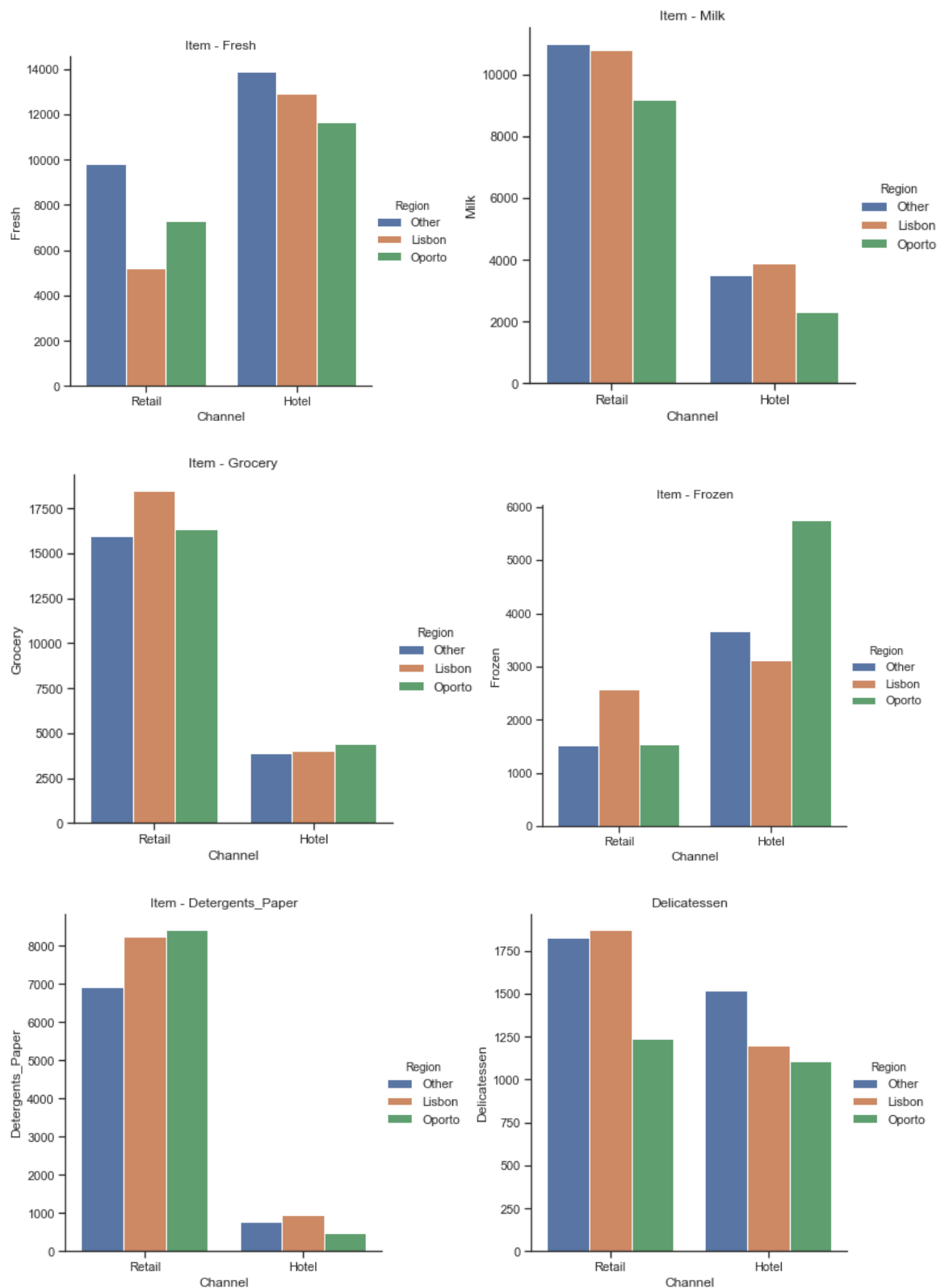
Channel
Hotel       7999569
Retail      6619931
Name: Spending, dtype: int64

```

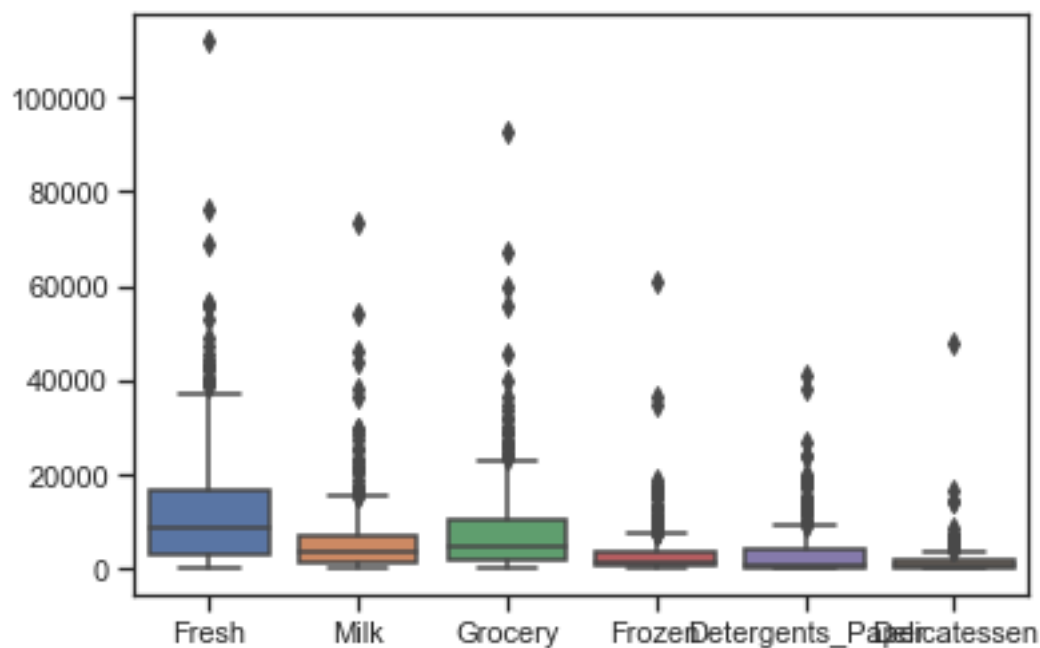
The Region that has spent the most is **Other (10677599)** and the region that has spent the least is **Oporto (1555088)**.

The Channel that has spent the most is **Hotel (7999569)** and the channel that has spent the least is **Retail(6619931)**.

1.2 There are 6 different varieties of items that are considered. Describe and comment/explain all the varieties across Region and Channel? Provide a detailed justification for your answer.



Looking at the above tables, we see that some categories like Milk, Grocery & Detergents Paper have higher spend in the Retail channel versus Hotel, across all regions. On the other hand, Fresh and Frozen have higher consumption in the Hotel channel versus Retail, across all regions. Also, if we plot a box plot, we can summarize that the spend for Fresh and groceries is the maximum across region and channel while for Delicatessen it is the least across region and channel. The output boxplot is below –



On the basis of above analysis, it can be concluded that considering all the 6 variety of items, all varieties do not show similar behaviour across Region and Channel

1.3 On the basis of a descriptive measure of variability, which item shows the most inconsistent behaviour?

Which items show the least inconsistent behaviour?

Based on Standard Deviation

Fresh	12647.33
Milk	7380.38
Grocery	9503.16
Frozen	4854.67
Detergents_Paper	4767.85
Delicatessen	2820.11
Spending	26356.30

dtype: float64

Fresh item has highest Standard deviation So that is **Inconsistent**.

Delicatessen item have smallest Standard deviation, so that is **consistent**.

Coefficient of Variance(CV) = μ/σ

where:

σ = standard deviation

μ = mean

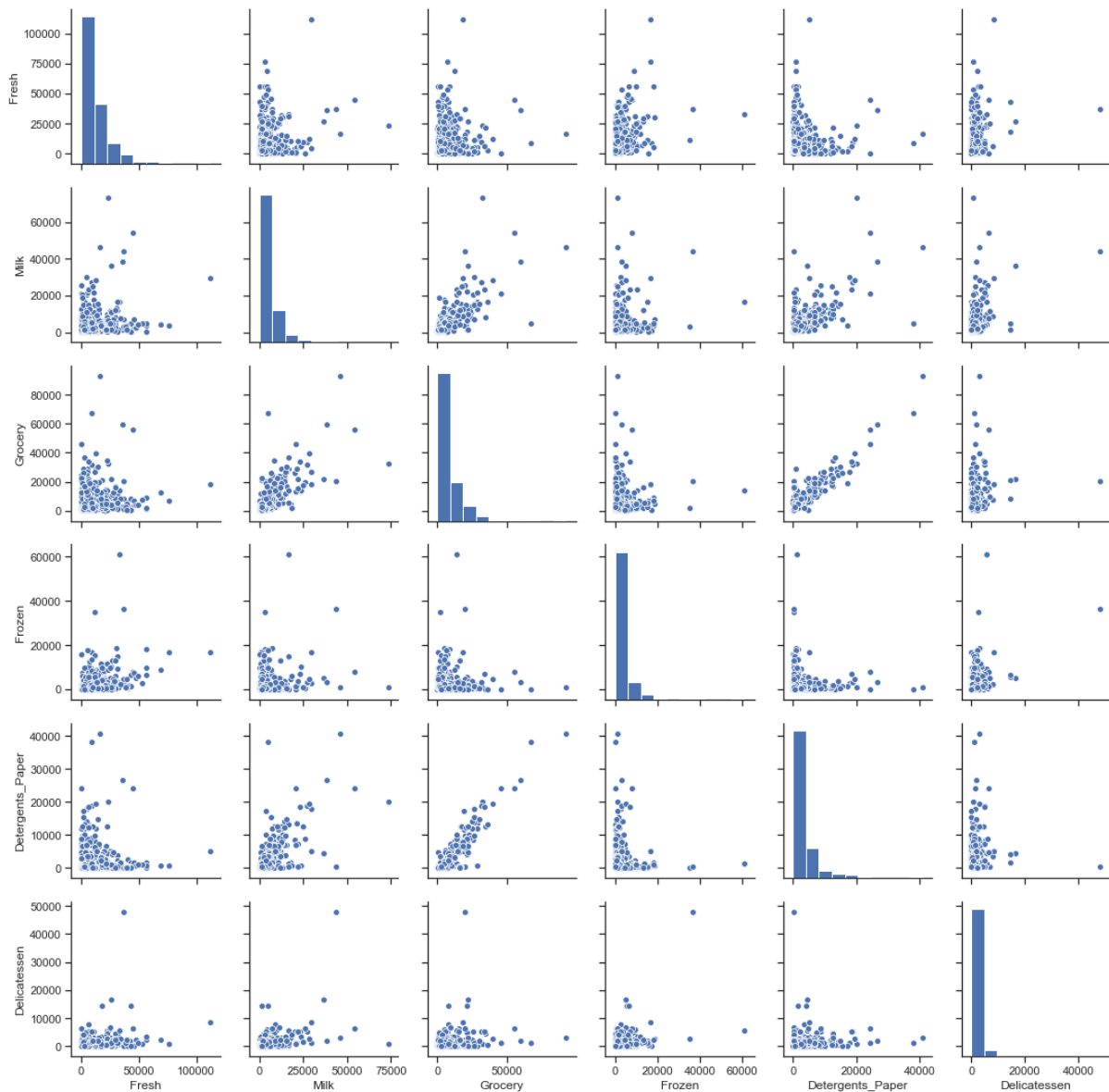
Based on Co-Efficient of Variation

Fresh	1.599549e+08
Milk	5.446997e+07
Grocery	9.031010e+07
Frozen	2.356785e+07
Detergents_Paper	2.273244e+07
Delicatessen	7.952997e+06
Spending	6.946546e+08

dtype: float64

Fresh item has lowest coefficient of Variation So that is **consistent**.

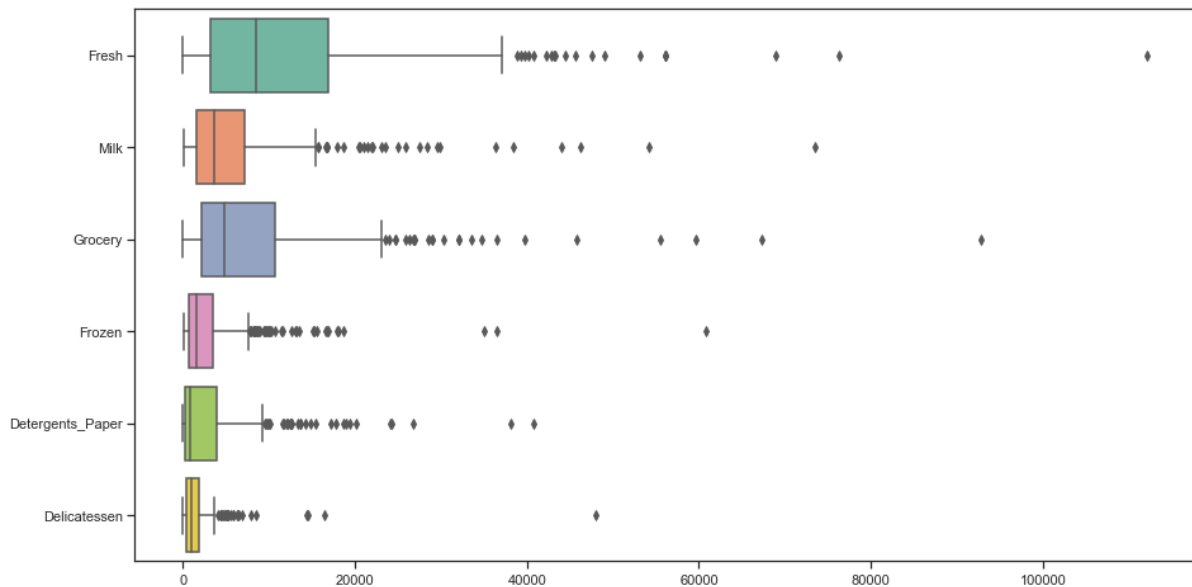
Delicatessen item has highest coefficient of Variation, So that is **inconsistent**.



This pair plot helps us to understand the relationship between the 6 food items.

1.4 Are there any outliers in the data? Back up your answer with a suitable plot/technique with the help of detailed comments.

To determine the presence of Outliers in the Data the best method is creating Box plot of all the variables as shown below.



From the Box plots of all the Variables as above it can be concluded that Yes, there are outliers in all the items across the product range (Fresh, Milk, Grocery, Frozen, Detergents Paper & Delicatessen)

1.5 On the basis of your analysis, what are your recommendations for the business? How can your analysis help the business to solve its problem?
Answer from the business perspective

On the basis of the analysis the following recommendations can be made:

- On the basis of the analysis, it can be seen that the region **Other** and the channel **Retail** have Higher spending than other Channel and Regions. Hence From the Business perspective if a new business is to be opened it Should be opened in the **other** region with Channel **Retail** as the Other region is absorbing maximum amount of sell and this can boast up the Revenue compared to opening a new business in Lisbon or Oporto and with the Channel Hotel.
- In all the regions the Foot Items **Fresh** has the highest spending followed by **Grocery** and **Milk**. Hence these food products are strongly recommended to be available simultaneously at all the businesses with priority of availability being **Fresh** food products.
- Also, the food item **Delicatessen** shows least inconsistent behaviour across all regions and channels. So, **Delicatessen** is also recommended to be available at all times in all the Businesses.

2 – Clear Mountain State University (CMSU) Survey

The Student News Service at Clear Mountain State University (CMSU) has decided to gather data about the undergraduate students that attend CMSU. CMSU creates and distributes a survey of 14 questions and receives responses from 62 undergraduates (stored in the **Survey** data set).

The Data is stored in the **Survey** data set as follows:

	ID	Gender	Age	Class	Major	Grad Intention	GPA	Employment	Salary	Social Networking	Satisfaction	Spending	Computer	Text Messages
0	1	Female	20	Junior	Other	Yes	2.9	Full-Time	50.0	1	3	350	Laptop	200
1	2	Male	23	Senior	Management	Yes	3.6	Part-Time	25.0	1	4	360	Laptop	50
2	3	Male	21	Junior	Other	Yes	2.5	Part-Time	45.0	2	4	600	Laptop	200
3	4	Male	21	Junior	CIS	Yes	2.5	Full-Time	40.0	4	6	600	Laptop	250
4	5	Male	23	Senior	Other	Undecided	2.8	Unemployed	40.0	2	4	500	Laptop	100
5	6	Female	22	Senior	Economics/Finance	Undecided	2.3	Unemployed	78.0	3	2	700	Laptop	30
6	7	Female	21	Junior	Other	Undecided	3.0	Part-Time	50.0	1	3	500	Laptop	50
7	8	Female	22	Senior	Other	Undecided	3.1	Full-Time	80.0	1	2	200	Tablet	300
8	9	Female	20	Junior	Management	Yes	3.6	Unemployed	30.0	0	4	500	Laptop	400
9	10	Female	21	Senior	Economics/Finance	Undecided	3.3	Part-Time	37.5	1	4	200	Laptop	100

2.1. For this data, construct the following contingency tables (Keep Gender as row variable)

2.1.1. Gender and Major

Major	Accounting	CIS	Economics/Finance	International Business	Management	Other	Retailing/Marketing	Undecided	
Gender									
Female		3	3	7	4	4	3	9	0
Male		4	1	4	2	6	4	5	3

2.1.2. Gender and Grad Intention

Grad Intention	No	Undecided	Yes
Gender			
Female	9	13	11
Male	3	9	17

2.1.3. Gender and Employment

Employment	Full-Time	Part-Time	Unemployed
Gender			
Female	3	24	6
Male	7	19	3

2.1.4. Gender and Computer

Computer	Desktop	Laptop	Tablet
Gender			
Female	2	29	2
Male	3	26	0

2.2. Assume that the sample is representative of the population of CMSU. Based on the data, answer the following question:

2.2.1. What is the probability that a randomly selected CMSU student will be male?

Total No of Students = 62

Total No of Male = 29

Probability of male student = number of male student/total number of students= $29/62 = 0.46774193548387094$

The probability that a randomly selected CMSU student will be male is **0.4677419354838** or **46.77%**

2.2.2. What is the probability that a randomly selected CMSU student will be female?

Total No of Students = 62

Total No of Female = 33

Probability of female student = number of female student /total number of students= $33/62 = 0.532258064516129$

The probability that a randomly selected CMSU student will be female is **0.53225806451** or **53.22%**

2.3. Assume that the sample is representative of the population of CMSU. Based on the data, answer the following question:

Major	Accounting	CIS	Economics/Finance	International Business	Management	Other	Retailing/Marketing	Undecided
Gender								
Female	0.090909	0.090909	0.212121	0.121212	0.121212	0.090909	0.272727	0.000000
Male	0.137931	0.034483	0.137931	0.068966	0.206897	0.137931	0.172414	0.103448
All	0.112903	0.064516	0.177419	0.096774	0.161290	0.112903	0.225806	0.048387

2.3.1. Find the conditional probability of different majors among the male students in CMSU.

From all the contingency tables created it can be seen that.

Probability of Accounting among the male students = $4/29$

Probability of CIS among the male students = $1/29$

Probability of Economics/Finance among the male students = $4/29$

Probability of International Business among the male students = $2/29$

Probability of Management among the male students Management = $6/29$

Probability of Other among the male students Other = $4/29$

Probability of Retailing/Marketing among the male students = $5/29$

Probability of Undecided among the male students = $3/29$

Hence from the calculations done in Python we conclude that :

The Probability of Accounting among the male students is 13.79%

The Probability of CIS among the male students is 3.45%

The Probability of Economics/Finance among the male students 13.79%

The Probability of International Business among the male students 6.9%

The Probability of Management among the male students Management is 20.69%

The Probability of Other among the male students Other 13.79%

The Probability of Retailing/Marketing among the male students 17.24%

The Probability of Undecided among the male students 10.34%

2.3.2 Find the conditional probability of different majors among the female students of CMSU.

From all the contingency tables created it can be seen that.

Probability of Accounting among the female students = $3/33$

Probability of CIS among the female students = $3/33$

Probability of Economics/Finance among the female students = $7/33$

Probability of International Business among the female students = $4/33$

Probability of Management among the female students Management = $4/33$

Probability of Other among the female students Other = $3/33$

Probability of Retailing/Marketing among the female students = $9/33$

Probability of Undecided among the female students = $0/33$

Hence from the calculations done in Python we conclude that :

The Probability of Accounting among the female students is 9.09%

The Probability of CIS among the female students is 9.09%

The Probability of Economics/Finance among the female students 21.21%

The Probability of International Business among the female students 12.12%

The Probability of Management among the female students Management is 12.12%

The Probability of Other among the female students Other 9.09%

The Probability of Retailing/Marketing among the female students 27.27%

The Probability of Undecided among the female students 0%

2.4. Assume that the sample is a representative of the population of CMSU. Based on the data, answer the following question:

2.4.1. Find the probability That a randomly chosen student is a male and intends to graduate.

Gender	Grad Intention	
Female	No	9
	Undecided	13
	Yes	11
Male	No	3
	Undecided	9
	Yes	17

Probability that a randomly chosen student is a Male = $29/62$

Probability of Male that intends to Graduate = $17/29$

Probability a randomly chosen student is a male and intends to graduate

= Probability that a randomly chosen student is a Male * Probability that a randomly chosen student is a Male

Hence from the calculations done in Python we conclude that :

The probability That a randomly chosen student is a male and intends to graduate is 27.42 %

2.4.2 Find the probability that a randomly selected student is a female and does NOT have a laptop.

Gender	Computer	
Female	Desktop	2
	Laptop	29
	Tablet	2
Male	Desktop	3
	Laptop	26

Probability that a randomly chosen student is a Female = $33/62$

Probability of Female with No Laptop = $1-(29/33)$

Probability that a randomly selected student is a female and does NOT have a laptop

= Probability that a randomly chosen student is a Female * Probability of Female with No Laptop

Hence from the calculations done in Python we conclude that :

The probability that a randomly selected student is a female and does NOT have a laptop is 6.45 %

2.5. Assume that the sample is representative of the population of CMSU. Based on the data, answer the following question:

2.5.1. Find the probability that a randomly chosen student is a male or has full-time employment?

Gender	Employment	
Female	Full-Time	3
	Part-Time	24
	Unemployed	6
Male	Full-Time	7
	Part-Time	19
	Unemployed	3

Probability of a Student being Male = $29/33$

Probability of a student having FullTime Employment = $10/62$

Probability of a Male having FullTime Employment = $7/29$

Probability that a randomly chosen student is either a male or has full-time employment

= Probability of a Student being Male + Probability of a student having FullTime Employment - Probability of a Male having FullTime Employment

Hence from the calculations done in Python we conclude that :

The probability that a randomly chosen student is either a male or has a full-time employment 79.87 %

2.5.2. Find the conditional probability that given a female student is randomly chosen, she is majoring in international business or management.

Gender	Major	
Female	Accounting	3
	CIS	3
	Economics/Finance	7
	International Business	4
	Management	4
	Other	3
	Retailing/Marketing	9

Probability of international business given Female = $4/33$

Probability of management given Female = $4/33$

Since international business and management are independent of each other

Probability of international business or management given Female

= Probability of international business given Female + Probability of management given Female

Hence from the calculations done in Python we conclude that :

The conditional probability that given a female student is randomly chosen, she is majoring in international business or management is 24.242 %

2.6. Construct a contingency table of Gender and Intent to Graduate at 2 levels (Yes/No). The Undecided students are not considered now and the table is a 2x2 table. Do you think the graduate intention and being female are independent events?

Gender	Grad Intention	
Female	No	9
	Undecided	13
	Yes	11
Male	No	3
	Undecided	9
	Yes	17

Two events A and B can be proved to be Independent events when it satisfies the condition :

$$P(A \cap B) = P(A) * P(B)$$

In this case if being female and graduate intention are independent can be proven by checking the condition :

$$P(F \cap \text{Yes}) = P(F) * P(\text{Yes})$$

Where F = Female

Yes = Grad Intention being Yes

Hence from the calculations done in Python we conclude that :

$$P(F \cap \text{Yes}) \neq P(F) * P(\text{Yes})$$

Hence, Graduate intention and being female are not independent events.

2.7. Note that there are four numerical (continuous) variables in the data set, GPA, Salary, Spending, and Text Messages.

Answer the following questions based on the data

2.7.1. If a student is chosen randomly, what is the probability that his/her GPA is less than 3?

Probability that a randomly chosen student's GPA is less than 3 = number of students with GPA less than 3/total number of students=17/62

Probability that a randomly chosen student's GPA is less than 3 is 0.27419354838709675

2.7.2. Find the conditional probability that a randomly selected male earns 50 or more. Find the conditional probability that a randomly selected female earns 50 or more.

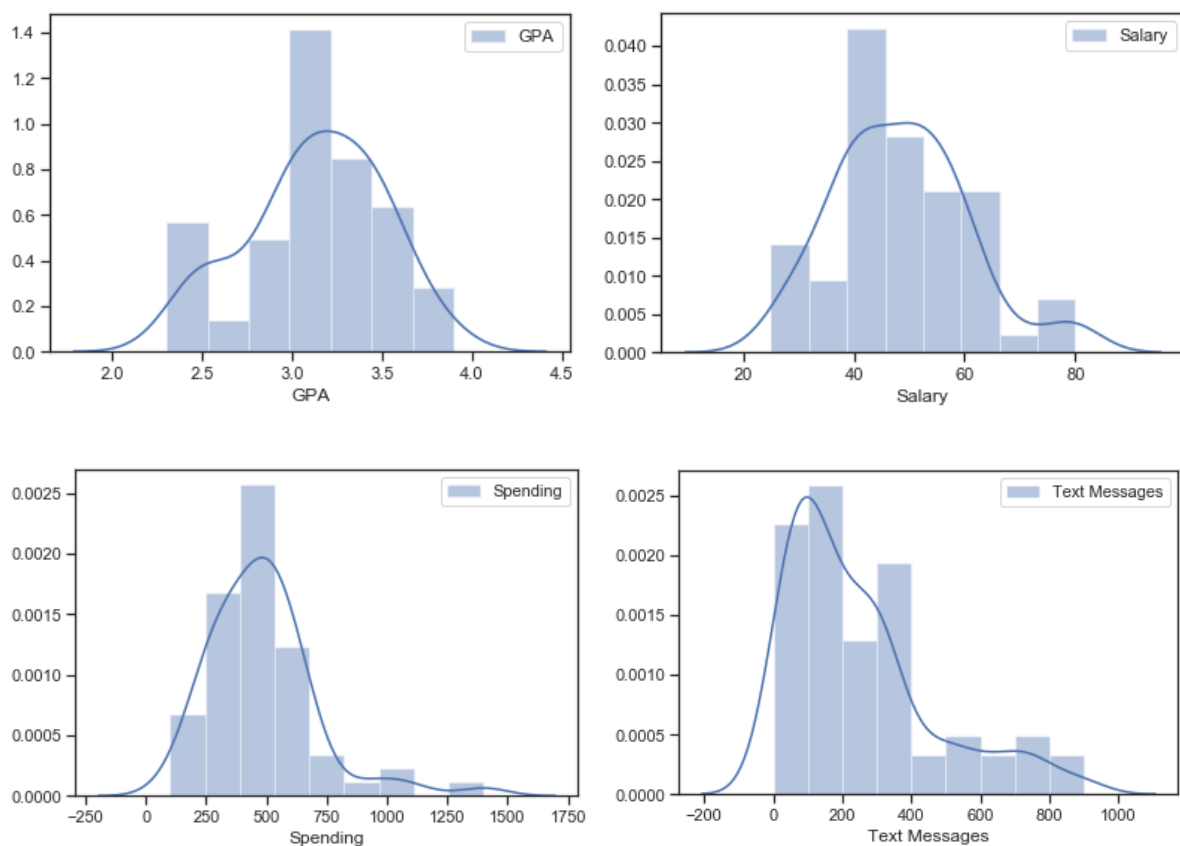
Probability that a randomly selected male earns more than 50 = number male students who earns 50 or more/total number of male=14/29

Probability that a randomly selected male earns more than 50 is 0.4827586206896552

Probability that a randomly selected Female earns more than 50 = number of females earning 50 or more/number of females=18/33 Probability that a randomly selected Female earns more than 50 is 0.5454545454545454

2.8. Note that there are four numerical (continuous) variables in the data set, GPA, Salary, Spending, and Text Messages. For each of them comment whether they follow a normal distribution. Write a note summarizing your conclusions.

Used distplot to know the normal distribution of these four numerical (continuous) variables in the data set – GPA, Salary, Spending and Text Messages



By these details we confirm that out of the given four data sets ‘GPA’ and ‘Salary’ are following normal distribution whereas other two ‘Spending’ and ‘Text Messages’ are not following the normal distribution

3 - Hypothesis Testing for Quality of Shingles

An important quality characteristic used by the manufacturers of ABC asphalt shingles is the amount of moisture the shingles contain when they are packaged. Customers may feel that they have purchased a product lacking in quality if they find moisture and wet shingles inside the packaging. In some cases, excessive moisture can cause the granules attached to the shingles for texture and coloring purposes to fall off the shingles resulting in appearance problems. To monitor the amount of moisture present, the company conducts moisture tests. A shingle is weighed and then dried. The shingle is then reweighed, and based on the amount of moisture taken out of the product, the pounds of moisture per 100 square feet are calculated. The company would like to show that the mean moisture content is less than 0.35 pounds per 100 square feet.

The file ([A & B shingles.csv](#)) includes 36 measurements (in pounds per 100 square feet) for A shingles and 31 for B shingles.

3.1 Do you think there is evidence that means moisture contents in both types of shingles are within the permissible limits? State your conclusions clearly showing all steps.

In this problem we have provided with two independent samples of shingles A and B population standard deviation is unknown and hence we can't perform z test. So we have to go with t-test.

Since we have to find the mean moisture level is less than the permissible limit for the both samples we have perform one sample t-test for sample A and sample B.

SAMPLE A

STEP 1:

DEFINE NULL AND ALTERNATE HYPOTHESIS

The null hypothesis states that the moisture content of sample A is greater or than equal to the permissible limit, $\mu \geq 0.35$

The alternative hypothesis states that the moisture content of sample A is less than permissible limit, $\mu < 0.35$

$$H_0 : \mu \geq 0.35$$

$$H : \mu < 0.35$$

STEP 2:

DECIDE THE SIGNIFICANCE LIMIT

Since alpha value is not given in the question we assume it has alpha = 0.05

STEP 3

IDENTIFY THE TEST STATISTIC

We have sample A and we do not know the population standard deviation. Sample size n=36.

We use the t distribution and the *tSTAT* test statistic for one sample t-test.

STEP 4:**CALCULATE THE P - VALUE AND TEST STATISTIC**

$$\bar{X} = 0.316667$$

$$S = 0.135731$$

$$N = 36$$

$$\mu = 0.35$$

$$T_{\text{stat}} = -1.4735$$

$$(P \text{ Value}/2) = 0.0747$$

STEP 5:**DECIDE TO REJECT OR ACCEPT NULL HYPOTHESIS**

Since $t_{\text{stat}} > p_{\text{value}}$, we fail to reject the null hypothesis

We conclude that the moisture content is greater than permissible limit in sample A.

SAMPLE B**STEP 1:****DEFINE NULL AND ALTERNATE HYPOTHESIS**

The null hypothesis states that the moisture content of sample B is greater or than equal to the permissible limit, $\mu \geq 0.35$

The alternative hypothesis states that the moisture content of sample B is less than permissible limit, $\mu < 0.35$

$$H_0 : \mu \geq 0.35$$

$$H : \mu < 0.35$$

STEP 2:**DECIDE THE SIGNIFICANCE LIMIT**

Since alpha value is not given in the question we assume it has $\alpha = 0.05$

STEP 3

IDENTIFY THE TEST STATISTIC

We have sample A and we do not know the population standard deviation. Sample size $n=31$.

We use the t distribution and the $tSTAT$ test statistic for one sample t-test.

STEP 4:

CALCULATE THE P - VALUE AND TEST STATISTIC

$$\bar{X} = 0.2735$$

$$S = 0.1372$$

$$N = 31$$

$$\mu = 0.35$$

$$T_{stat} = -3.1003$$

$$P \text{ Value} = 0.0020$$

STEP 5:

DECIDE TO REJECT OR ACCEPT NULL HYPOTHESIS

Since $t_{stat} < p_value$, we reject the null hypothesis

We conclude that the moisture content is less than permissible limit in sample B

3.2 Do you think that the population mean for shingles A and B are equal? Form the hypothesis and conduct the test of the hypothesis. What assumption do you need to check before the test for equality of means is performed?

STEP 1

DEFINE NULL AND ALTERNATIVE HYPOTHESIS

In testing whether the mean for shingles A and Shingles B are the same, the null hypothesis states that

the mean of shingle A to mean of shingle B are the same, $\mu\{A\}$ equals $\mu\{B\}$. The

alternative hypothesis states that the mean are different, $\mu\{A\}$ is not equal to $\mu\{B\}$

STEP 2:

DECIDE THE SIGNIFICANCE LIMIT

Since alpha value is not given in the question we assume it has $\alpha = 0.05$

STEP 3

IDENTIFY THE TEST STATISTIC

We have two samples and we do not know the population standard deviation.

Sample sizes for both samples are not the same.

The sample size is, $n > 30$. So we use the t distribution and the *tSTAT* test statistic for two sample test.

Two tail test

STEP 4:

CALCULATE THE P - VALUE AND TEST STATISTIC

Tstat 1.2896282719661123

P Value 0.2017496571835306

STEP 5

DECIDE TO REJECT OR ACCEPT THE NULL HYPOTHESIS

Since $tstat > p_value$, we fail to reject the null hypothesis

We conclude that mean for shingles A and singles B are not the same.